
REQUIREMENTS NOT MET

N/A

PROBLEMS ENCOUNTER'ED

N/A

FUTURE WORK/APPLICATIONS

This lab prepares us for the standards of labs going forward to ensure a fair, smooth process for every student.
Doing this lab prepares us to do well in the course (so we can all get A's 😊)

PRE-LAB EXERCISES

- i. Describe the lab makeup policy for a first missed (i.e., not submitted) lab.

If you never submit a lab, that would count as a drop; however, you are allowed to submit the assignment late up to 24 hours, along with a 25 point deduction to your score.

- ii. You cannot be late for your lab appointment time, but you can turn in your lab submissions late. How late, what are the penalties, and if so, how many times.

Students can turn in at most **two** labs up to 24 hours late for a 25-point penalty. (Third and after get 0 credit.)

- iii. What can I do if I miss my lab appointment, but I already submitted the lab document? What is the grade penalty?

If a student submits the document but misses their lab appointment, students can reach out to their PI within 24 hours to reschedule a **single** lab appointment based on a mutually agreeable time.

- iv. Can you drop any (single) lab if you would rather go skiing or to the beach? How about if a project for another class is due?

A student does get a single lab drop to use as they see fit, however, they will only get one free drop. If a student needs to miss a second lab for any reason, they must reach out to Professor Schwartz with proper official documentation on reasons for dropping **BOTH** labs.

- v. Other than Labs 0, when are your pre-lab submissions due to Canvas, with respect to the earliest corresponding lab demo dates and lab quiz, i.e., how many days before or after the earliest lab or the quiz? Be specific.

This can vary, as most labs are submitted between Sunday and Tuesday (Lab 4 being an outlier on Thursday, 8 on Wednesday.) But basically, we will submit our labs **typically between 2-3 days ahead** of time of our lab quizzes and for me as a Tuesday lab, about 1-2 days ahead of time of my lab demos, although there may be some weeks where this differs.

- vi. In your pre-lab report, what should be included with every measurement, screenshot, etc.?

Captions for every measurement.

- vii. What is the minimum combined weighted lab average and lab quiz average required for you to be eligible to pass the course?

65% weighted average between labs and lab quizzes.

- viii. In which section of your pre-lab report should any required screenshot be included?

Within the appendix section.

- ix. Describe the process of taking measurements with a system such as an oscilloscope or logic analyzer. Additionally, include details regarding when precise frequency measurements are of interest.

We will use the Waveforms from Diligent, with our scope 1 and 2 probes, and our I/O pins. For Scopes, we will use a common ground for the negative and probe the pin of interest as necessary.

For logic analyzers, we will use I/O pins, and connect our inputs and output, inputting a waveform generator pin utilizing the patterns section from the Waveforms software. This will show you the I/O for your current setup, and help you solve the logic behind it.

For frequency data, we will again use the scope, and do a horizontal measurement for period and frequency information (avoiding using traces as that can show false data in unstable signals.)

PSEUDOCODE/FLOWCHARTS

PROGRAM CODE

APPENDIX
